

To evaluate equal opportunity, we assess whether self-preference persists when resumes are of equivalent quality, as defined in Eq. (2). We use two complementary approaches: (1) a conditional logistic regression controlling for content quality features, and (2) human annotations indicating which resume is objectively better.

As shown in Figure 3b, after adjusting for content quality using conditional logistic regression, most evaluator models continue to exhibit significant self-preferencing behavior. Consistent with the statistical parity results, all models except LLaMA-3.2-1B display positive and significant self-preference bias. Larger models—such as GPT-4o, GPT-4-turbo, DeepSeek-V3, Qwen-2.5-72B, and LLaMA-3.3-70B—exhibit the strongest LLM-vs-Human self-preference, with magnitudes ranging from 67% to 82%. In contrast, smaller models display weaker biases: Mistral-7B and LLaMA-3.2-3B show modest but statistically significant effects (28% and 11.6%, respectively). More detailed conditional logistic regression results from Eq. (3) are presented in Appendix F (Table EC.1).

We further assess equal opportunity using human-annotated ground truth. Strikingly, across all three evaluator LLMs with human-annotated comparisons (GPT-4o, DeepSeek-V3, and LLaMA 3.3-70B), each model consistently selected its own generated summary over the human-written alternative—even in cases where human annotators judged the human-written summary to be higher quality in its presentation of the same underlying candidate (e.g., clearer, more coherent, or more effective overall).

5.2. LLM-vs-LLM Self-Preference

We next examine whether self-preference extends to cases where LLMs evaluate content generated by alternative LLMs. Specifically, we focus on GPT-4o, DeepSeek-V3, and LLaMA 3.3-70B, three of the most advanced and widely adopted models in both commercial and research contexts. These models not only achieve state-of-the-art performance across benchmarks (Hurst et al. 2024, DeepSeek-AI 2024, Dubey et al. 2024) but also represent diverse development lineages (OpenAI, DeepSeek, and Meta, respectively), making them especially relevant for understanding how self-preference may manifest across different model families and design philosophies. In this setting, each model serves as the evaluator LLM and is presented with pairwise comparisons between two counterfactual resumes: one generated by itself and the other by a competing LLM. Both resumes are derived from the same underlying human-written original, ensuring comparability of content. In contrast to the consistent self-preferencing bias documented in LLM-vs-Human comparisons, LLM-vs-LLM self-preferencing behavior proves considerably more heterogeneous across models.

5.2.1. Statistical Parity Self-Preference Bias

Analogous to the LLM-vs-Human setting, we assess statistical parity by comparing the probability that an evaluator LLM selects its own generated resume with the probability that it selects

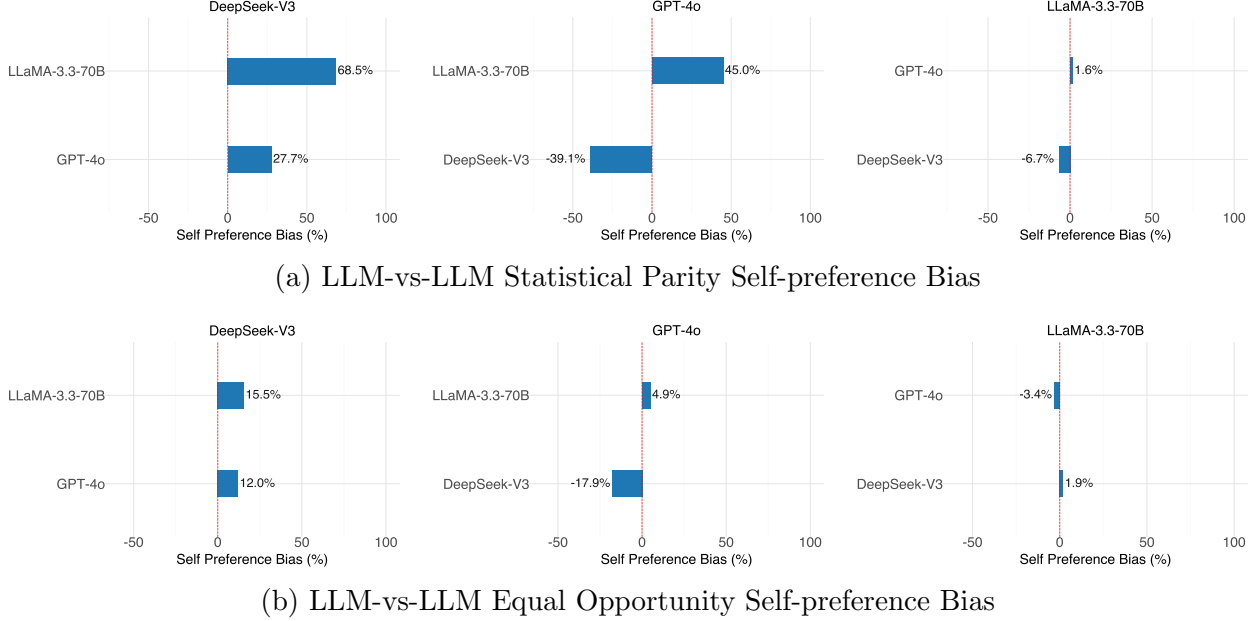


Figure 4 LLM-vs-LLM Self-Preference Bias

a resume produced by an alternative LLM, as defined in Eq. (1). As shown in Figure 4a, self-preferencing across LLMs is heterogeneous. DeepSeek-V3 is 69% more likely to select its own resumes than those generated by LLaMA-3.3-70B, and 28% more likely than those generated by GPT-4o. GPT-4o is 45% more likely to select its own resumes than those produced by LLaMA-3.3-70B, but this tendency reverses against DeepSeek-V3, where it is 39% more likely to select DeepSeek-V3’s resumes over its own. LLaMA-3.3-70B shows only weak asymmetries: it is 1.6% more likely to select its own resumes over GPT-4o’s, but 6.7% more likely to select DeepSeek-V3’s resumes over its own.

5.2.2. Equal Opportunity Self-Preference Bias

Similar to the LLM-vs-Human setting, we apply two complementary approaches to evaluate equal opportunity. First, we estimate conditional logistic regressions that control for content quality features. As shown in Figure 4b, DeepSeek-V3 exhibits the strongest and most consistent self-preference bias among the three models, showing biases of 15.5% against LLaMA 3.3-70B and 12% against GPT-4o. By contrast, GPT-4o does not exhibit strong self-preference relative to LLaMA 3.3-70B and, in fact, shows a negative bias when compared with DeepSeek-V3, favoring DeepSeek-V3’s outputs over its own. LLaMA 3.3-70B shows little self-preference in either comparison. Detailed conditional logistic regression results are provided in Table EC.2.

Second, we benchmark model decisions against human judgments of resume quality. These annotation-based results provide weaker evidence of equal opportunity self-preference. Although LLaMA 3.3-70B has a positive estimated bias of 12.94% relative to GPT-4o, the estimate is not

statistically significant (95% CI: $[-21.89\%, 47.27\%]$). Similarly, neither DeepSeek-V3 nor GPT-4o exhibits statistically significant equal opportunity self-preference under the annotation-based benchmark, despite showing strong statistical parity self-preference bias in the previous analysis.

These findings highlight that LLM-vs-LLM self-preference is model-specific and generally weaker than LLM-vs-Human self-preference. The heterogeneity observed suggests that self-preference is not a uniform property of LLMs but may instead reflect model-specific factors, such as differences in stylistic alignment or the ability to recognize patterns in their own outputs.

5.3. Robustness Checks

In the following, we assess the robustness of our empirical results by considering an alternative approach to constructing counterfactual resumes in the correspondence experiment. In our baseline design, counterfactual summaries are generated from the remainder of the resume content. As a robustness check, we instead prompt each LLM to *revise* the existing human-written executive summary by improving its clarity and wording, while explicitly preserving the candidate’s experience and qualifications. This alternative construction mirrors a common and realistic use case of AI-assisted writing, in which job applicants employ LLMs to polish their resumes rather than generate summaries from scratch. This design allows us to examine whether AI self-preference persists under AI-assisted revision rather than reconstruction, thereby ruling out the possibility that our main results are driven by the specific method used to generate counterfactual text. Using this revised setup, we re-estimate both LLM-vs-Human and LLM-vs-LLM self-preference biases under the same evaluation framework, including conditional logistic regressions to assess equal opportunity bias.

As shown in Figure 5a and Figure 5b, across both statistical parity and equal opportunity metrics, the qualitative patterns of LLM-vs-Human self-preference remain largely unchanged under the resume revision design. In both specifications, most evaluator LLMs continue to favor summaries revised by themselves over the original human-written summaries. Although the magnitude of self-preference is attenuated, for some models the magnitude still exceed 90%. Such results suggest that LLM-vs-Human self-preference is a robust phenomenon that arises even when LLM-generated text is obtained through revision rather than generation from scratch. The corresponding conditional logistic regression estimates are reported in Table EC.3.

Turning to LLM-vs-LLM self-preference, the revision-based design yields more heterogeneous estimates than the baseline reconstruction specification (see Figure 6a and Figure 6b), but several qualitative patterns persist. Consistent with the conditional logistic regression results reported in Table EC.4, DeepSeek-V3 continues to exhibit the strongest and most consistent equal-opportunity self-preference. Under the revision condition, DeepSeek-V3 displays equal-opportunity biases of